



Loop a Longer Pattern

Quick facilitation notes & answer key

Goal

Students see that a loop runs every block inside it each pass, so a multi-action pattern repeats as a unit. They count each action across the repeats by multiplying, and work backwards from a total.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app (Scratch-style) to project; the printed worksheet + a pencil per student; optional: do the clap-clap-stomp beat as a class.

Run it (about 30 min)

1. Hook: perform a clap-clap-stomp beat as a class; name the group that repeats.
2. Project the repeat-4 (clap, clap, stomp) loop; count each action together.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: multiply per-repeat actions by the repeat count.

Answer key (checked by the run-gate)

Ex 1 — 3 actions per repeat; 12 actions total.

Ex 2 — 5 stomps and 10 claps.

Ex 3 — repeat 4; 8 waves total.

Ex 4 (challenge) — 6 repeats (6 stomps ÷ 1 stomp per repeat).

Watch for & differentiate

Common slip: counting the pattern once instead of multiplying by the repeats.

Support: circle one action type and skip-count it across the repeats.

Extend: ask how many claps in repeat 5 of (clap, clap, stomp) — 10.

Success looks like

The student counts actions per repeat and in total (Ex 1), predicts a single action's count after changing the repeat (Ex 2), writes a pattern loop (Ex 3), and works backwards from a total to the repeat count (Ex 4).